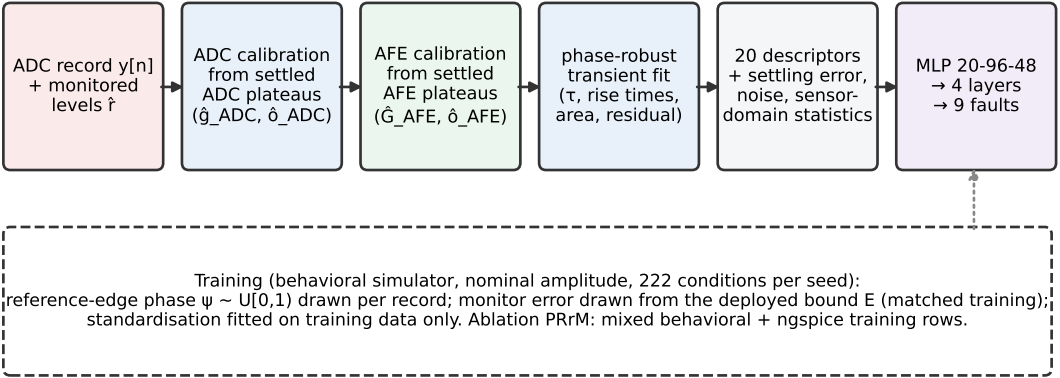
Processing pipeline of the proposed phase-robust, reference-monitored self-test
ADC record y[n] + monitored levels r̂
ADC calibration from settled ADC plateaus (ĝ_ADC, ô_ADC)
AFE calibration from settled AFE plateaus (Ĝ_AFE, ô_AFE)
phase-robust transient fit (τ, rise times, area, residual)
20 descriptors + settling error, noise, sensor-domain statistics
MLP 20-96-48 → 4 layers → 9 faults
Training (behavioral simulator, nominal amplitude, 222 conditions per seed):
reference-edge phase ψ ~ U[0,1) drawn per record; monitor error drawn from the deployed bound E (matched training);
standardisation fitted on training data only. Ablation PRrM: mixed behavioral + ngspice training rows.